Weight-Loss Study Randomization Test

MATH 213

Weight Loss Study

- As in the Sex Discrimination study, random assignment was used here, so confounding variables *should be* minimized and it *should be* the case that the two groups are similar in terms of all characteristics.

- But how can we become more sure that the difference in mean weight loss for the groups is actually caused by the incentive and is not due to the inherent variability of the random assignment process?

  i.e. How do we know that the random assignment process didn't accidentally put all of the people who would otherwise have lost the most weight into the same group?

- Like last time, we will use a randomization test to check this.

Weight Loss Randomization Test

Just like in the Sex Discrimination study, we consider two competing claims: a null hypothesis and an alternative hypothesis. The null hypothesis is always the hypothesis of no effect.

Null hypothesis, H_0: The incentive had no effect. Subjects would have lost the same amount of weight regardless of which group they were in.

Alternative hypothesis, H_a: The incentive had an effect. Being in the incentive group *caused* the subjects to lose weight.

Setting up the Randomization Test

- To start the randomization test, we ASSUME the null hypothesis and then check to see how unusual the actual experimental result seems under these conditions.

- That is, we will assume that the subjects in the study would have lost the same amount of weight no matter what group they were in.

Discussion Questions:

If we were going to do a playing card simulation for this randomization test:

1. What would the cards represent?
2. How many cards would there be?
3. Last time, we had red cards and blue cards to represent supervisors who “promote” and “don’t promote”. What would we have to write on the cards this time?
4. After I shuffle (and shuffle, and shuffle), how would I deal out the cards?

Answers for Discussion Questions

If we were going to do a playing card simulation for this randomization test:

1. What would the cards represent? Each card will represent a study participant.
2. How many cards would there be? There will be 36 cards, one for each person in the study.
3. Last time, we had red cards and blue cards to represent supervisors who “promote” and “don’t promote”. What would we have to write on the cards this time? This time, red and blue have no role to play. Instead, it would be better to use blank index cards instead of playing cards. On each card (representing a subject), I would write the amount of weight that subject lost by the 16-week mark.
4. After I shuffle (and shuffle, and shuffle), how would I deal out the cards?

After we shuffle, because we are assuming (null hypothesis) that people would have lost the same amount of weight regardless of what group they were in, we deal out the cards randomly into two groups: 19 cards for the control group, and 17 for the incentive group.

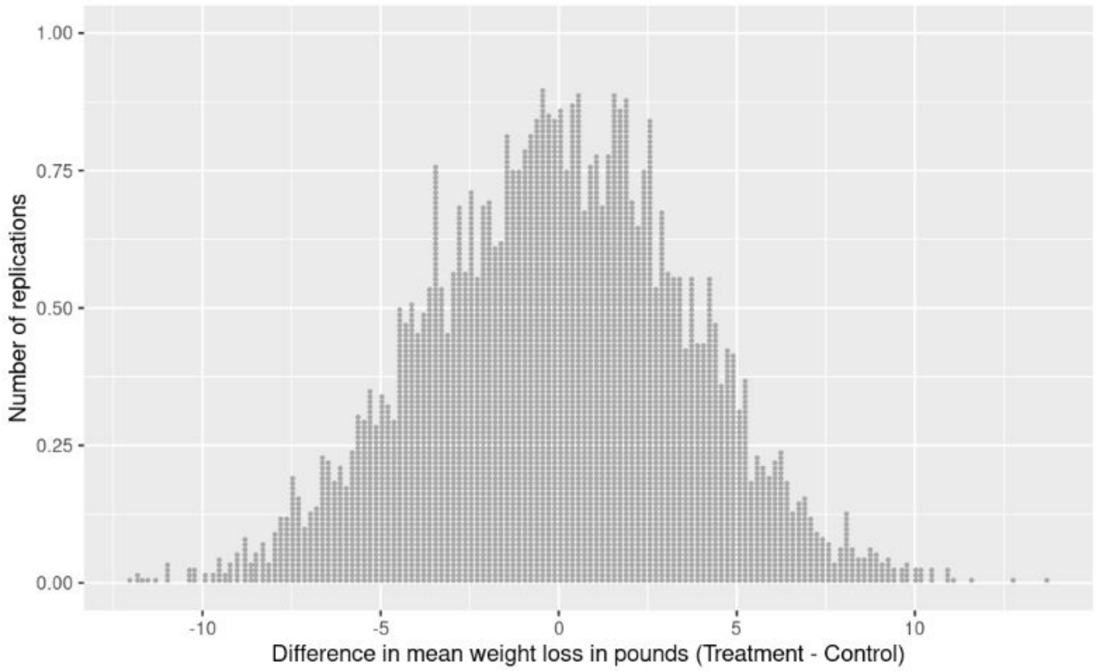
1. Then, compute the mean weight loss in each group, subtract them incentive - control, and plot that difference on a dotplot.
2. Repeat the simulation lots and lots of times. Judge how unusual the experimental difference of 11:755lbs seems.

Results of the randomization test in R

We had R run a simulation that was the equivalent of shuffling the index cards, dealing them into a “control” group and an “incentive” group (also called the “treatment” group in the graph). Then, R calculated the mean weight loss for each group and subtracted those values in the order Treatment - Control and we were able to plot the dots in a dotplot. We had R do this 5000 times to create 5000 dots in the dotplot.

This dotplot gives us a sense of what the differences in mean weight loss would look like under the assumption of the null hypothesis that the treatment had no effect on the participants’ ability to lose weight.

In the actual ***experiment***, the difference in mean weight loss between the groups was 11.755 lbs. This seems pretty unusual compared to what we see in the dotplot. To quantify just how unusual of a result this would be under the null hypothesis, I worked out that we got a difference greater than or equal to 11.755 lbs appearing in the dotplot only TWICE out of our 5000 simulations. Those are the two little dots all the way to the right of the dotplot in the picture.



See the next slide for a paragraph outlining the conclusions we can draw from this randomization test.

Paragraph: Conclusions from this randomization test

Assuming that the incentive plan had no effect on the participants’ amount of weight loss, the difference in mean weight loss of 11.75lbs between those in the treatment (incentive) group and those in the control group from the experiment seems quite unusual. In fact, a result as or more extreme than this occurred only 2 times out of 5000 in our simulation in R. This means that the probability of getting a difference of means as or more extreme than the experimental result of 11.75lbs was 2/5000 = 0.0004. (This number is called the **p-value** for the hypothesis test.) The probability of this occurring is so small, that it seems extremely unlikely that the weight loss plan did not have an effect on the amount of weight lost.